

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 33)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

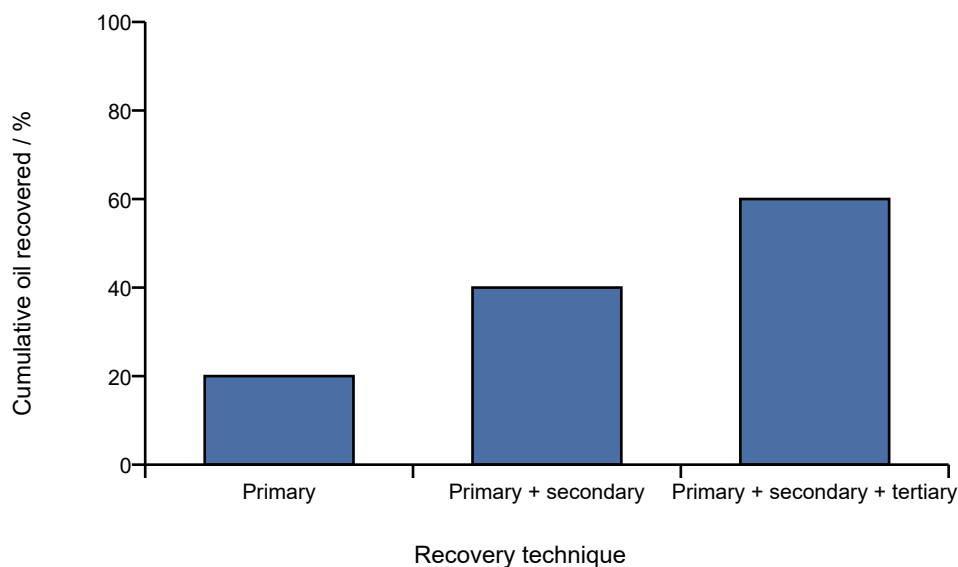
Question	Mark
1	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 1

1

Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 1

2

Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 1

3

Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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4

 Explain **three** advantages of directional drilling.

[3 marks]

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0 2

Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO₂ over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NO _x)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NO _x and hydrocarbons	Days-weeks	-

0 2**1**

Explain the natural greenhouse effect.

[3 marks]

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0 2**2**

Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 2 . 3

Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO₂ emissions by the same mass.

[2 marks]

0 2 . 4

Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

1

2

10

0 3 . 1

Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

0 3 . 2

Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

0 3 . 3

Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

10

0 4

Table 1 shows three specialist solid wastes.

0 4 . 1

Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

0	4	.	2
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Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0	4	.	3
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Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

0	4	.	4
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Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

10

0 5

A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 5**1**

Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 5**2**

Give **three** limitations of kick sampling.

[3 marks]

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0 5**3**

Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 5**4**

Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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10

0 6 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

0 6 . **1** Complete Table 1 by explaining how each feature reduces oil pollution. **[5 marks]**

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

0 6 . **2** Explain how each of the following reduces the risk of tanker accidents. **[2 marks]**

Offshore shipping routes

GPS / Automatic Identification System

0 6 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank. **[2 marks]**

0 6 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park. **[1 mark]**

10

0 7 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 7 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland. **[1 mark]**

Answer t/ha

07 . 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

07 . 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

2

5

08

Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg^{-1}	Cadmium in kidney of residents / mg kg^{-1}	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

08 . 1

Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer

times

08 . 2

Explain why cadmium and zinc pollution often occur together.

[1 mark]

08 . 3

Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

08 . 4

Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

1

2

08 . 5

Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

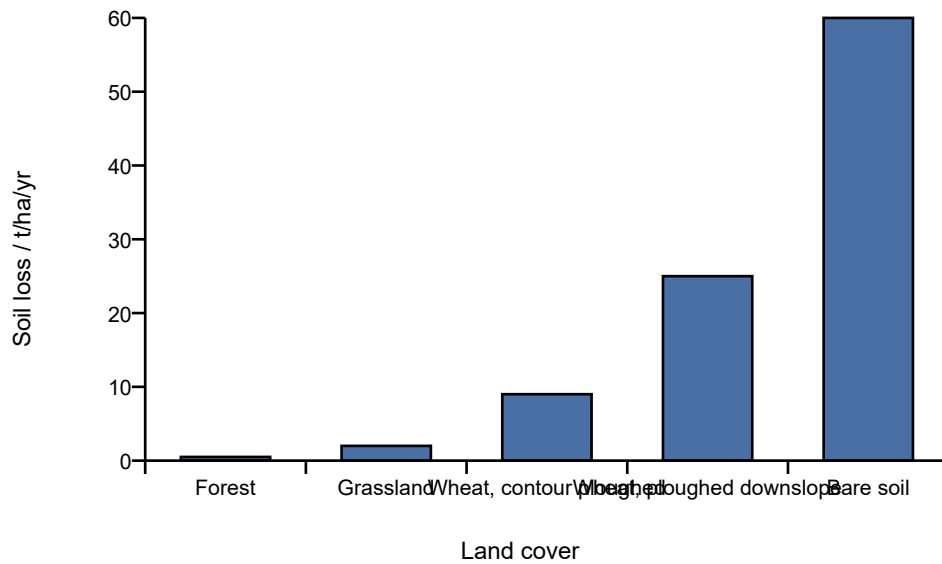
[2 marks]

10

09

Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1



09

1

Describe the pattern shown in Figure 1.

[2 marks]

09

2

Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

0	9	.	3
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Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

0	9	.	4
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Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

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Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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10 . 2

State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2

Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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END OF QUESTIONS